CUDA Programming Interface

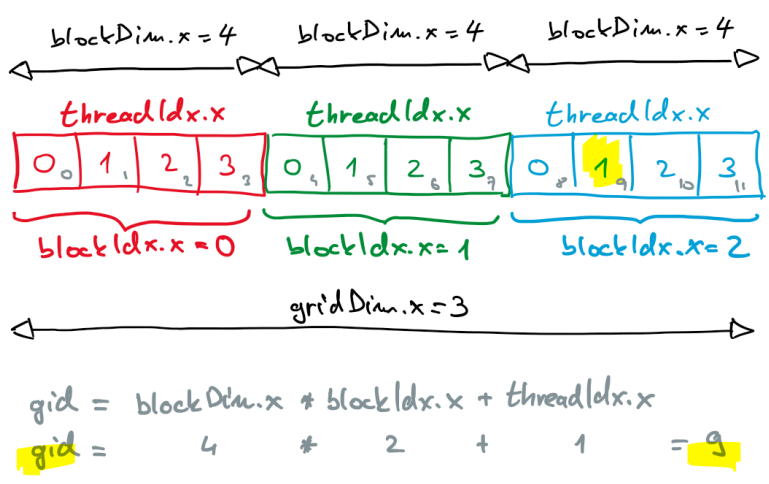
- a CUDA kernel represents code that runs on a GPU
- the kernel is written as a sequential program
- the programming interface handles the compilation and transfer of the kernel to the device
- the kernel executes individually on its own data for each thread

Hierarchical Organization of Threads

- a grid of threads
  - all threads in the grid execute the same kernel
  - threads in the grid share global memory on the GPU
  - the grid consists of thread blocks
- a thread block
  - all threads in a block are executed on the same compute unit
  - they can exchange and synchronize through local (shared) memory
- a thread warp
  - group of consecutive threads that follows SIMT (single instruction multiple threads principle)
  - 32 threads at Nvidia GPUs and 64 at AMD GPUs
- a thread
  - executes the kernel sequentially on its own data
  - uses its private memory
  - shares the local memory with other threads in the block
  - can access global memory and constant memory

Thread Indexing

- 1D, 2D, or 3D indexing
- threads are grouped in thread warps first by dimension x, then y, and finally z
- the number of dimensions is chosen based on the nature of the problem
- CUDA C supports a set of variables reflecting thread organization:
  - `threadIdx.x`, `threadIdx.y`, `threadIdx.z`
  - `blockIdx.x`, `blockIdx.y`, `blockIdx.z`
  - `blockDim.x`, `blockDim.y`, `blockDim.z`
  - `gridDim.x`, `gridDim.y`, `gridDim.z`
- thread indexing



GPU Kernel

- a GPU kernel is code that is started from the host but runs on the device
- the kernel is written as a function, prefixed by the keyword `__global__`
- the kernel does not return a value
- a kernel example:

```
__global__ void greetings(void) {
  printf("Hello from thread %d.%d!\n", blockIdx.x, threadIdx.x);
}
```

- the kernel is launched on the host, where triple angle brackets are inserted between the name and arguments
- thread organization in the grid - the number of blocks and the number of threads in each dimension - is specified within the triple angle brackets
- for describing multidimensional thread organization, the CUDA C language provides the `dim3` structure:

```
dim3 gridSize(numBlocks, 1, 1);
dim3 blockSize(numThreads, 1, 1);
greetings<<<gridSize, blockSize>>>();
```

- a kernel can also call other device functions marked with the `__device__` keyword
- to emphasize that a function runs only on the host, it is marked with `__host__`

The first GPU program

- hello-gpu.cu
- load the module: `module load CUDA`
- compile the code with the CUDA C compiler: `nvcc -o hello-gpu hello-gpu.cu`
- run the program: `srun --partition=gpu --gpus=1 ./hello-gpu 2 4`

Memory Allocation and Data Transfer

- the host has access only to the global memory of the device

Explicit Data Transfer

- on the host, memory is allocated using the `malloc` function
- global memory on the device is allocated using the function call:

```
cudaError_t cudaMalloc(void** dPtr, size_t count)
```

- this function allocates `count` bytes and returns the address in the device's global memory to the pointer `dPtr`
- to transfer data between the device's global memory and the host memory, we use the function:

```
cudaError_t cudaMemcpy(void* dst, const void* src, size_t count,
         cudaMemcpyKind kind)
```

- this function copies `count` bytes from the address `src` to the address `dst` in the direction specified by kind, which is
  - for transferring data from host to device `cudaMemcpyHostToDevice`, and
  - for transferring data from device to host `cudaMemcpyDeviceToHost`
- the function is blocking - the program execution continues only after the data transfer is complete
- device memory is freed with the function call:

```
cudaError_t cudaFree(void *devPtr)
```

- memory on the host is freed using function `free`

Unified Memory

- newer versions of CUDA support unified memory
- CUDA performs data transfers as needed
- a programmer has no control, and it is often less efficient than explicit transfers
- the unified memory is allocated using the function call:

```
cudaError_t cudaMallocManaged(void **hdPtr, size_t count);
```

- the unified memory is freed using the function call:

```
cudaError_t cudaFree(void *hdPtr)
```

- in our examples we will not work with the unified memory

Example: SAXPY

- Single precision A times X Plus Y
- vectors **x** and **y**
- element-wise operation `y[i] = a * x[i] + y[i]`
- the map pattern
- solutions
  - saxpy0.cu: support for one thread block
  - saxpy1.cu: added support for multiple thread blocks, the number of blocks is calculated based on the problem size
  - saxpy2.cu: improved code, threads with global index out of the vector (array) bounds don't do any work
  - saxpy3.cu: in case when total number of threads is smaller than the problem size, some threads do additional work
  - saxpy4.cu: unified memory